

Project 4: Preference elicitation

Human-Centric Artificial Intelligence

In this project, you will design a user study to compare two methods for eliciting user preferences. As for Project 3, **we are expecting a PDF report**, containing a complete description of the user study and of your design choices. The report must be accessible from the project interface (for instance with a download button). **You are not expected to conduct the user study.** The goal of this project is to design a rigorous experimental protocol and implement the corresponding interactive interface that could be used to run the study.

1 Dataset and Methods

1.1 Dataset

The goal of this project is to design a movie recommender that can quickly adapt to the preferences of a new user. You will use the IMDB 5000 Movie Dataset. This dataset contains metadata describing approximately 5000 movies, including various attributes. Unlike traditional recommender datasets, it does *not* contain user ratings or preference data.

1.2 Preference model

For a given movie and user, we define a utility function $U(x) = w^T x$, where x is the vector of features of the movie and w is the user's latent preference vector. The purpose of the preference elicitation process is to estimate w from a limited number of interactions with the user.

Task 1

Choose a feature representation that you believe is appropriate for a movie recommender system. Justify your choice of features and write a method to extract them from the dataset.

We assume that user preferences follow a Bradley–Terry model. In its standard formulation, the Bradley-Terry model defines the probability that one item is preferred to another based on their respective utilities.

In this project, however, one of the interfaces will ask participants to rank a list of movies rather than compare only two movies. You will therefore need to extend the model to account for rankings involving multiple items.

Task 2

Propose an extension of the Bradley-Terry model capable of modeling a ranking of $i_1 \succ i_2 \succ \dots \succ i_n$ instead of a single pairwise comparison. Explain and justify your proposed formulation.

2 User study

The objective of the user study is to compare two different preference elicitation interfaces.

- **Design 1:** the participant is repeatedly presented with two movies and chooses the one they would rather watch.
- **Design 2:** the participant is presented with a set of ten movies and is asked to rank them from the most preferred to the least preferred.

In both designs, participants perform a sequence of preference elicitation tasks. The movies presented to the participant may be selected uniformly at random from the dataset. (An interesting extension would be to investigate more informative or adaptive selection strategies.)

Task 3

Design a complete user study comparing the two elicitation methods. You have to choose the hypothesis you are testing, the design of the user study, the recruitment process etc. Explain the steps that you would follow in practice if you would have to really run the user study. **(I remind you that you don't have to run the user study!)**

Task 4

Implement the interface for the user study. This interface is the one that would be given to a participant to the study.

In terms of organization of the project page, create a landing page, from which (1) I can download a PDF providing explanations about the implemented method (Tasks 1 and 2) and the design of the user study (Task 3); and (2) I can start the study itself (Task 4).